A Survey of Deep Learning-Based Reconstruction Methods for Quantum Imaging Systems

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Abstract

Quantum imaging systems—including ghost imaging, entangled-photon compressed sensing, and sub-shot-noise microscopy—exploit non-classical photon correlations to surpass classical limits in sensitivity and resolution. However, reconstructing high-quality images from the sparse, noisy measurements intrinsic to these systems remains computationally demanding. This survey systematically reviews deep learning architectures that have been applied to quantum imaging reconstruction. We categorise methods along three axes: imaging modality, neural network architecture, and reconstruction task. Our review reveals that deep learning enables measurement reductions of one to two orders of magnitude in ghost imaging, identifies a convergence towards U-Net and GAN-based architectures, and highlights critical gaps in standardised benchmarking. We conclude by discussing hybrid classical–quantum reconstruction frameworks and proposing directions for quantum-inspired neural architectures in imaging pipelines.

Keywords: quantum imaging, ghost imaging, deep learning, compressed sensing, image reconstruction, quantum optics, neural networks

1. Introduction

Quantum imaging harnesses the non-classical correlations of entangled or correlated photon pairs to achieve imaging modalities fundamentally inaccessible to classical light sources. Since the seminal demonstration of imaging via two-photon entanglement by Pittman et al. [1995], the field has expanded to encompass ghost imaging, quantum illumination, sub-shot-noise microscopy, and entanglement-assisted compressed sensing. These techniques promise advantages in spatial resolution, noise robustness, and the ability to image with photons that never interact with the object [Erkmen and Shapiro, 2010].

Despite these remarkable capabilities, a persistent bottleneck remains: *image reconstruction*. Quantum imaging measurements are inherently sparse—limited by detector efficiencies, coincidence counting rates, and photon budgets—and the resulting inverse problems are severely ill-posed. Traditional reconstruction approaches rely on correlation computation or iterative optimisation, which demand large numbers of measurements and significant computation time.

The emergence of deep learning as a powerful framework for solving inverse problems in computational imaging [Barbastathis et al., 2019, LeCun et al., 2015] has opened new avenues for quantum imaging reconstruction. Neural networks can learn strong image priors from data, enabling high-quality reconstruction from far fewer measurements than classical algorithms require. This intersection of quantum optics and deep learning has produced a rapidly growing body of work spanning multiple imaging modalities and network architectures.

1.1 Scope and Contributions

This survey provides a systematic review of deep learning methods applied to quantum imaging reconstruction. We focus on three principal modalities:

(i) **Quantum ghost imaging**, where deep learning dramatically reduces measurement requirements for image reconstruction from spatially correlated photon fields.

(ii) **Compressed sensing with quantum correlations**, where neural networks enhance reconstruction from sub-Nyquist quantum measurements.

(iii) **Sub-shot-noise imaging**, where deep learning denoising exploits quantum noise statistics to push sensitivity below the classical shot-noise limit.

The contributions of this survey are:

- A unified taxonomic framework classifying methods by imaging modality, DL architecture, and reconstruction task.

- A detailed comparative analysis of deep learning approaches for ghost imaging reconstruction, including quantitative performance comparisons.
- Identification of critical gaps in benchmarking, reproducibility, and standardisation.
- A discussion of emerging hybrid classical–quantum architectures and future research directions.

1.2 Paper Organisation

Section 2 reviews the necessary background in quantum imaging and deep learning for inverse problems. Section 3 surveys deep learning methods for ghost imaging in detail. Section 4 covers compressed sensing with quantum correlations. Section 5 addresses sub-shot-noise imaging. Section 6 examines hybrid classical–quantum architectures. Section 7 discusses open problems, and Section 8 concludes.

2. Background

2.1 Quantum Imaging Fundamentals

Quantum imaging relies on photon pairs generated through spontaneous parametric down-conversion (SPDC), in which a pump photon incident on a nonlinear crystal produces two daughter photons—conventionally termed *signal* and *idler*—that are correlated in position, momentum, and time. These correlations arise from the phase-matching conditions of the nonlinear interaction and can be either classical or genuinely quantum (entangled) depending on the experimental configuration.

The spatial correlations between signal and idler photons enable imaging protocols where information about an object is extracted by measuring correlations between spatially separated detectors, rather than by direct imaging. This principle underlies ghost imaging, quantum illumination, and related protocols.

2.2 Ghost Imaging: From Quantum to Computational

Ghost imaging (GI) was first demonstrated by Pittman et al. [1995] using entangled photon pairs from SPDC. In the original configuration, the object is placed in the path of one photon (say, the signal), which is detected by a single-pixel (bucket) detector that records no spatial information. The partner idler photon, which never interacts with the object, is detected by a spatially resolving detector. Individually, neither detector's output contains an image; the image emerges only from the *coincidence* correlations between the two detectors.

The quantum nature of the original ghost imaging was debated after Bennink et al. [2002] demonstrated a similar effect using classically correlated thermal light, and Ferri et al. [2005] achieved high-resolution ghost imaging with pseudothermal sources. Shapiro [2008] subsequently introduced *computational* ghost imaging (CGI), in which the spatially resolving detector is replaced entirely by computational knowledge of the illumination patterns. In CGI, a spatial light modulator generates known random patterns, the object is illuminated by each pattern in sequence, and a bucket detector records the total transmitted intensity. The image is reconstructed by correlating the known patterns with the bucket detector readings:

$$I(\mathbf{r}) = \frac{1}{M} \sum_{m=1}^{M} (B_m - \langle B \rangle)\, \phi_m(\mathbf{r}), \tag{1}$$

where B_m is the bucket detector signal for the m-th pattern $\phi_m(\mathbf{r})$, $\langle B \rangle$ is the mean bucket signal, and M is the total number of measurements. High-quality reconstruction via Eq. (1) typically requires M to be comparable to or greater than the number of pixels in the reconstructed image [Erkmen and Shapiro, 2010], motivating the development of more efficient reconstruction methods.

2.3 Deep Learning for Inverse Problems

Deep learning has emerged as a dominant paradigm for solving inverse problems in imaging, where the goal is to recover a signal $\mathbf{x}$ from measurements $\mathbf{y} = \mathcal{A}(\mathbf{x}) + \mathbf{n}$, with $\mathcal{A}$ denoting the forward measurement operator and $\mathbf{n}$ representing noise [Barbastathis et al., 2019].

Several architecture families are particularly relevant to quantum imaging. Convolutional neural networks (CNNs) learn hierarchical spatial features through convolutional filters, enabling effective image-to-image mappings for denoising and reconstruction [LeCun et al., 2015]. The U-Net architecture [Ronneberger et al., 2015] extends the convolutional approach with an encoder–decoder structure and skip connections that preserve fine-grained spatial information, and has been widely adopted for biomedical and computational imaging tasks. Generative adversarial networks (GANs) employ a generator–discriminator framework that produces perceptually realistic reconstructions by learning the distribution of natural images [Goodfellow et al., 2014]. Autoencoders, which learn compressed latent representations from which images can be reconstructed, are also relevant for dimensionality reduction in quantum measurement data.

Barbastathis et al. [2019] provide a comprehensive review of deep learning for computational imaging, noting that learned approaches can outperform traditional iterative methods in both speed and quality, particularly when training data adequately represents the target image distribution.

2.4 Compressed Sensing Theory

Compressed sensing (CS) provides a mathematical framework for recovering sparse signals from sub-Nyquist measurements [Donoho, 2006, Candès and

Tao, 2006]. A signal $\mathbf{x} \in \mathbb{R}^n$ that is k-sparse in some basis Ψ can be recovered from $m \ll n$ linear measurements $\mathbf{y} = \Phi\mathbf{x}$ if the measurement matrix Φ satisfies the restricted isometry property (RIP).

The relevance of CS to quantum imaging is twofold. First, quantum correlations can provide natural incoherence between measurement and sparsity bases, satisfying CS recovery conditions [Katz et al., 2009]. Second, the extreme photon scarcity in quantum imaging makes sub-Nyquist reconstruction practically essential. Katz et al. [2009] demonstrated compressive ghost imaging, achieving image reconstruction with far fewer measurements than the Nyquist limit by exploiting image sparsity, and subsequent work has combined CS with deep learning to further improve reconstruction quality.

3. Deep Learning for Ghost Imaging Reconstruction

The application of deep learning to ghost imaging reconstruction has been the most active area at the intersection of quantum imaging and neural networks. This section reviews the principal approaches, organised by network architecture.

3.1 Fully Connected and CNN-Based Approaches

Lyu et al. [2017] provided the first demonstration that a neural network could reconstruct ghost images from bucket detector measurements. Their approach used a fully connected network trained on pairs of bucket signals and ground-truth images, achieving recognisable reconstruction of 28×28 handwritten digits at a sampling ratio as low as 12.5%—a dramatic improvement over the correlation-based method (Eq. 1), which requires sampling ratios approaching 100% for comparable quality.

He et al. [2018] extended this work with a convolutional architecture, demonstrating improved generalisation to object categories not seen during training. Their CNN employed multiple convolutional layers with batch normalisation and ReLU activations, and was validated on both simulated and experimental ghost imaging data. Notably, the CNN-based approach maintained reasonable reconstruction quality even at sampling ratios below 10%, where traditional methods produce severely degraded images.

Shimobaba et al. [2018] applied deep learning specifically to computational ghost imaging, replacing the correlation computation of Eq. (1) with a learned mapping. Their results confirmed that deep learning consistently outperforms traditional correlation-based reconstruction, particularly in low-measurement regimes.

3.2 GAN-Based Reconstruction

Generative adversarial networks [Goodfellow et al., 2014] have been applied to ghost imaging to improve the perceptual quality of reconstructions. The adversarial training framework encourages the generator to produce images that are statistically indistinguishable from real images, yielding reconstructions with sharper edges and more realistic textures than those obtained from MSE-trained networks.

However, GAN-based methods carry an inherent risk of *hallucination*—generating plausible but incorrect image features—which is particularly concerning in scientific imaging applications where fidelity to the true object is paramount. This trade-off between perceptual quality and pixel-level accuracy remains an active area of investigation.

3.3 End-to-End Learned Approaches

A significant advance was made by Wang et al. [2019], who proposed jointly optimising the illumination patterns and the reconstruction network in an end-to-end differentiable framework. Rather than using random illumination patterns and learning only the reconstruction mapping, their approach learns *optimal measurement strategies* tailored to the reconstruction network.

This end-to-end optimisation achieved 3–5 dB improvement in PSNR over methods that separately optimise illumination and reconstruction, demonstrating that the measurement and reconstruction stages are fundamentally coupled and benefit from joint design. The approach also provides insight into what constitutes an "informative" measurement in the context of ghost imaging.

3.4 Comparative Analysis

Table 1 summarises the key characteristics and reported performance of the reviewed deep learning methods for ghost imaging reconstruction.

Several trends emerge from this comparison. First, all methods achieve usable reconstruction quality at sampling ratios well below the Nyquist limit, confirming the strong regularising effect of learned image priors. Second, the field has progressed from simple fully connected architectures to more sophisticated end-to-end frameworks. Third, there is a notable lack of standardised benchmarks: different papers use different image sizes, datasets, noise models, and evaluation metrics, making rigorous cross-method comparison difficult.

3.5 Discussion

The reviewed works establish that deep learning is a transformative tool for ghost imaging reconstruction. However, several limitations persist. Most studies operate on small image sizes (28×28 to 64×64); scalability to high-resolution imaging remains underexplored.

Table 1: Comparison of deep learning methods for ghost imaging reconstruction.

Reference	Architecture	Size	Sampling	Data	Key Contribution
Lyu et al. [2017]	Fully connected	28×28	12.5%	Sim.	First DL-based ghost imaging reconstruction
He et al. [2018]	CNN	28×28	<10%	Sim.+Exp.	Improved generalisation to unseen categories
Shimobaba et al. [2018]	CNN	64×64	25%	Sim.	Applied DL to computational GI
Wang et al. [2019]	End-to-end CNN	64×64	6.25%	Sim.	Joint optimisation of patterns and network

The reliance on simulated data in many studies raises questions about robustness to real experimental noise and detector imperfections. Finally, the fundamental question of whether quantum correlations provide an advantage over classical correlations *when combined with deep learning* has not been systematically addressed.

4. Compressed Sensing with Quantum Correlations

Compressed sensing (CS) provides a natural framework for quantum imaging, where measurements are inherently limited by photon budgets and detector constraints. This section reviews the integration of deep learning with CS approaches that exploit quantum correlations.

4.1 Quantum Compressed Sensing Fundamentals

Katz et al. [2009] demonstrated that ghost imaging can be formulated as a compressed sensing problem, achieving image reconstruction from significantly fewer measurements than the Nyquist limit by exploiting the sparsity of natural images. In this framework, the bucket detector measurements constitute the compressed observations $\mathbf{y} = \Phi\mathbf{x}$, where Φ is determined by the illumination patterns and $\mathbf{x}$ is the image to be recovered.

The quantum advantage in this context arises from two sources. First, entangled photon pairs can provide measurement matrices with favourable incoherence properties, naturally satisfying the conditions required for CS recovery [Donoho, 2006, Candès and Tao, 2006]. Second, quantum correlations enable noise rejection through coincidence detection, effectively improving the signal-to-noise ratio of each measurement.

4.2 Neural Network Enhanced CS Reconstruction

Traditional CS recovery algorithms (e.g., basis pursuit, iterative hard thresholding) are computationally expensive and require careful tuning of regularisation parameters. Deep learning offers two strategies for improvement: (i) *purely data-driven* approaches that learn a direct mapping from compressed measurements to images, bypassing explicit sparsity assumptions; and (ii) *deep unfolding* approaches that parameterise iterative CS algorithms as neural networks, combining the interpretability of model-based methods with the flexibility of learned parameters.

Both strategies have shown promise in classical compressed imaging and are beginning to be explored in the quantum setting, where the unique noise statistics of photon-counting measurements introduce additional challenges.

4.3 Entanglement-Assisted Measurement Optimisation

An emerging direction explores the use of entangled photon states to design measurement bases that are optimally suited for compressed sensing recovery. This connects to the broader field of quantum state tomography, where similar compressed measurement strategies have been developed. The potential for deep learning to jointly optimise entangled measurement strategies and reconstruction networks—extending the end-to-end approach of Wang et al. [2019] to the quantum domain—remains largely unexplored.

5. Sub-Shot-Noise Imaging

Sub-shot-noise imaging exploits quantum correlations between photon pairs to achieve sensitivity below the classical shot-noise limit, which bounds the minimum noise in measurements made with coherent (classical) light sources.

5.1 Quantum Noise Limits in Imaging

Brida et al. [2010] provided the first experimental demonstration of sub-shot-noise quantum imaging, achieving a noise reduction of approximately 40% below the shot-noise level using spatially entangled photon pairs from SPDC. Samantaray et al. [2017] subsequently realised the first sub-shot-noise wide-field microscope, extending the technique from proof-of-concept to a practical imaging modality capable of operating over extended fields of view.

The quantum enhancement in these systems arises from the photon-number correlations between signal and idler beams: by subtracting the idler intensity from the signal, classical noise sources are suppressed below the shot-noise floor. However, the improvement is limited by optical losses and detector inefficiencies, which degrade the quantum correlations.

5.2 Deep Learning Denoising for Quantum-Enhanced SNR

Deep learning denoising represents a promising direction for extending the practical advantage of sub-shot-noise imaging. Classical denoising algorithms assume Gaussian or Poissonian noise statistics, which do not accurately describe the sub-Poissonian noise in quantum-correlated imaging. Neural networks trained on data with appropriate quantum noise models could exploit the specific statistical structure of quantum measurements to achieve superior denoising performance.

Furthermore, deep learning may help compensate for the optical losses that currently limit practical sub-shot-noise imaging, effectively widening the parameter regime in which quantum advantage is achievable. This possibility has been explored in related contexts such as deep learning microscopy [Rivenson et al., 2017], but dedicated studies in the quantum imaging setting remain sparse.

6. Hybrid Classical–Quantum Architectures

An emerging frontier in quantum imaging reconstruction is the development of hybrid architectures that combine classical neural network components with quantum-inspired or quantum-computational elements.

6.1 Quantum-Inspired Neural Networks

Beer et al. [2020] demonstrated that deep quantum neural networks can be trained for supervised learning tasks, establishing the feasibility of parameterised quantum circuits as function approximators. While their work focused on general learning tasks rather than imaging, the architectures may be adaptable to quantum image reconstruction, particularly for problems where the data is natively quantum.

6.2 Variational Quantum Circuits for Image Processing

Killoran et al. [2019] introduced continuous-variable quantum neural networks operating on photonic quantum states, providing a framework that is naturally suited to quantum optical imaging systems. The potential for these architectures to process quantum imaging data without requiring intermediate classical conversion—thereby preserving quantum information—is theoretically appealing but experimentally challenging.

6.3 Integration Challenges and Opportunities

The practical integration of quantum computational elements into imaging pipelines faces significant challenges, including the limited qubit counts and gate fidelities of current noisy intermediate-scale quantum (NISQ) devices, the overhead of quantum–classical data conversion, and the difficulty of training parameterised quantum circuits. Nevertheless, the long-term potential for native quantum data processing motivates continued investigation.

7. Discussion and Open Problems

The preceding sections reveal several cross-cutting themes and open problems that merit discussion.

7.1 Benchmarking and Reproducibility

A critical gap in the current literature is the absence of standardised benchmarks for deep learning-based quantum imaging reconstruction. Different studies employ different image sizes, datasets, noise models, sampling ratios, and evaluation metrics, precluding rigorous cross-method comparison. The development of community benchmark datasets—ideally incorporating both simulated and experimental quantum imaging data—would significantly accelerate progress.

7.2 Scalability

Most reviewed deep learning methods operate on small images (28×28 to 64×64 pixels). Scaling to the high-resolution images required for practical applications introduces challenges in both network architecture design and training data generation. The computational cost of simulating quantum imaging systems at scale further compounds this limitation.

7.3 Quantum Advantage in the Presence of Deep Learning

A fundamental open question is whether quantum correlations provide a genuine advantage over classical correlations when both are combined with state-of-the-art deep learning reconstruction. If neural networks can compensate for the additional noise in classical imaging, the practical quantum advantage may be smaller than previously assumed. Systematic studies comparing quantum and classical imaging under identical DL reconstruction pipelines are needed.

7.4 Hardware–Software Co-Design

The end-to-end optimisation paradigm demonstrated by Wang et al. [2019] for ghost imaging suggests a broader opportunity for hardware–software co-design, where the optical system and reconstruction algorithm are jointly optimised. Extending this approach to quantum optical systems—where the measurement apparatus includes nonlinear crystals, parametric sources, and single-photon detectors—is a promising but technically challenging direction.

8. Conclusion

This survey has reviewed the emerging field of deep learning-based reconstruction for quantum imaging systems, covering ghost imaging, compressed sensing with quantum correlations, sub-shot-noise imaging, and hybrid classical–quantum architectures.

Our principal findings are as follows. First, deep learning enables measurement reductions of one to two orders of magnitude in ghost imaging reconstruction, with end-to-end learned approaches providing the largest gains through joint optimisation of measurement and reconstruction. Second, the field is converging towards U-Net and GAN-based architectures, mirroring trends in classical computational imaging. Third, critical gaps remain in standardised benchmarking, high-resolution scalability, and systematic evaluation of quantum advantage in the presence of deep learning.

The intersection of quantum imaging and deep learning remains a fertile area for research, with significant potential for both fundamental insights and practical applications. As quantum optical hardware matures and deep learning methods continue to advance, their synergistic integration promises imaging capabilities that transcend the limits of either approach alone.

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